HIMANSHU YADAV

Data Analyst | Python, SQL, Machine Learning | B.Tech Computer Science and Engineering
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SUMMARY

B.Tech Computer Science and Engineering student at IIIT Raichur (graduating 2027) with hands-on experience in data analysis, feature engineering, statistical evaluation, and predictive modelling using Python, SQL, Pandas, NumPy, and Scikit-Learn. Completed 7+ data-driven projects across healthcare, satellite imagery, time-series, and video analytics, each evaluated with clear metrics (accuracy, F1, AUC-ROC, IoU, PSNR, SSIM). Also built an LLM-powered research agent that retrieves evidence from documents and produces citation-backed reports. Strong foundation in data structures, algorithms, DBMS, and problem solving. Seeking a Data Analyst internship to turn data into actionable insights.

EDUCATION

IIIT Raichur – B.Tech Computer Science and Engineering CGPA: 7.1/10 Aug 2023 – May 2027
Relevant coursework: Data Structures and Algorithms, Machine Learning, Deep Learning, DBMS, Computer Vision, Operating Systems, OOP, Computer Networks

TECHNICAL SKILLS

Data Analysis: Pandas, NumPy, SciPy, Exploratory Data Analysis, Data Cleaning, Feature Engineering, PCA, Cross-Validation
Databases and Visualization: SQL, DBMS concepts (coursework and lab teaching); Matplotlib, Seaborn, TensorBoard
Machine Learning for Analytics: Scikit-Learn, XGBoost, LightGBM, Classification, Anomaly Detection, Time-Series Modelling (LSTM)
Model and Metric Evaluation: Accuracy, F1-Score, AUC-ROC, IoU, PSNR, SSIM, Cross-Validation
Deep Learning: PyTorch, TensorFlow, Keras, CNN, RNN, LSTM, Transfer Learning
Generative AI and NLP: LLM APIs, Retrieval-Augmented Generation, Embeddings, Semantic Search, Text Classification, TF-IDF
Programming: Python, SQL, C, C++, JavaScript
Tools: Jupyter Notebook, Google Colab, Git, FastAPI, Docker, Linux

PROJECTS

Drug Side Effect Prediction and Image Forgery Detection Python, XGBoost, Scikit-Learn, RDKit, TensorFlow 2024–2025
- Built a predictive model on the SIDER dataset using engineered molecular fingerprint features, reaching 87% accuracy, 0.82 F1, and 0.89 AUC-ROC with XGBoost and neural networks; compared models on F1 and AUC-ROC.
- Developed a multi-task CNN with Error Level Analysis on the CASIA dataset, achieving 92% binary accuracy and 0.78 IoU for segmentation.

Privacy-Preserving Federated Learning for Healthcare Python, Pandas, TensorFlow Federated 2025
- Processed and prepared healthcare data with Pandas and built a federated averaging framework with formal epsilon-delta differential privacy guarantees, keeping model accuracy within 2% of the centralized baseline.

Privacy-Preserving Video Analytics with Anomaly Detection Python, TensorFlow Federated, CNN, LSTM 2025
- Built a distributed video analytics system combining CNN and LSTM with Federated Learning and Differential Privacy, reaching accuracy within 3% of the centralized baseline.
- Implemented Gaussian-noise differential privacy and real-time anomaly detection with automated alert generation.

AURA – LLM-Powered Research Agent Python, LLM APIs, FastAPI, Docker 2026
- Built planner, researcher, and critic agents that retrieve evidence from documents and generate citation-backed reports, using text chunking, keyword search, and embedding-based semantic search, served via a FastAPI backend in Docker.

Satellite Image Analysis using Deep Learning TensorFlow, Keras, OpenCV 2025
- Applied SRCNN and EDSR-inspired CNN models for 4x upscaling of Sentinel-2 and Landsat imagery; measured with PSNR and SSIM (3–5 dB PSNR improvement, SSIM of 0.92).

Time-Series Model Compression (TinyML) TensorFlow Lite, PyTorch, ONNX 2025
- Compressed an LSTM time-series model 4x using INT8 quantization, structured pruning, and knowledge distillation with under 2% accuracy loss; deployed on edge hardware at 30+ FPS.

GridPINN – Physics-Informed Neural Network for AC Optimal Power Flow PyTorch, pandapower, CVXPY In progress
- Developing a physics-informed neural network to predict generator dispatch and bus voltages, with power-balance and constraint terms in the loss.
- Benchmarking against a standard neural-network baseline, DC-OPF (CVXPY), and exact AC-OPF solutions (pandapower), and measuring constraint violations.

EXPERIENCE

Machine Learning Researcher *IIIT Raichur* 2023 – Present
- Built 7+ data-driven projects in Python, TensorFlow, PyTorch, and Scikit-Learn, covering data preparation, feature engineering, model training, and metric-based evaluation.
- Deployed optimized models on NVIDIA Jetson Nano and MCU targets via INT8 quantization, structured pruning, and distillation.

Teaching Assistant – Engineering Mathematics, DBMS, OS, Java OOP Labs *IIIT Raichur* 2024 – Present
- Designed and evaluated lab assignments for 60+ students covering databases, process scheduling, and object-oriented programming.

ACHIEVEMENTS

- **Reliance Foundation Scholar 2023** – Merit-based national scholarship for academic excellence in technology.
- **GIAN Summer Course Participant** – IIT Indore, Summer 2025: Ministry of Education program on Solving Linear Systems using Recurrent Neural Networks.